

GPU PyFR PASS 6 – Finite Field and Benchmark MAE Mapping for NASA Hump

OpenAI Codex workflow inside `nasa-hump2`

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1 Objective

GPU PyFR PASS 6 has one core objective: obtain a hump solution that is finite enough to support an honest benchmark mapping. PASS 5 already proved that the previous PyFR exports were not just awkward to postprocess; they were globally non-finite. PASS 6 therefore focuses on numerical stability first, then on honest C_p extraction, and only then on benchmark MAE.

2 What PASS 5 Proved

PASS 5 established three important facts:

- the PASS 4 wall-extraction failure was a field-validity failure, not merely a plotting failure;
- switching from `ac-navier-stokes` to dimensional `navier-stokes` was worth testing;
- even the stabilized PASS 5 diagnostic still needed a direct field audit before any benchmark or wall metric could be trusted.

3 PASS 5 Diagnostic Job Status

See `PASS5_DIAGNOSTIC_STATUS.md`. The short version is:

- job 10706975 completed on Gilbreth,
- it wrote a VTU and pseudo-stat file,
- but the exported `Density`, `Velocity`, and `Pressure` arrays were still globally NaN,
- the pseudo-stat file first became non-finite at pseudo iteration 4 of the first physical step.

4 Stabilization Sweep

PASS 6 responds with a small, focused sweep rather than a broad search. The sweep tests:

1. a rectangular smoke case to see if the dimensional `navier-stokes` path can stay finite at all,
2. two hump cases with successively smaller time steps,
3. two hump cases that replace the subsonic inflow/outflow pair with characteristic boundaries.

Case	dt	final time	finite vel-x	Cp MAE	benchmark MAE
pass6_hump_ns_char_dt1e4	0.0001	0.0005	316770	1.5251664939519705	0.29634149006117916
pass6_hump_ns_char_dt5e5	5e-05	0.00025	316770	1.1768448193661394	0.29371445045471833
pass6_hump_ns_sub_dt1e4	0.0001	0.0005	316770	1.4896397545412254	0.2963414500446999
pass6_hump_ns_sub_dt5e5	5e-05	0.00025	316770	1.1673277827939506	0.2937144506169717
pass6_smoke_ns_sub	0.0002	0.001	None	None	None

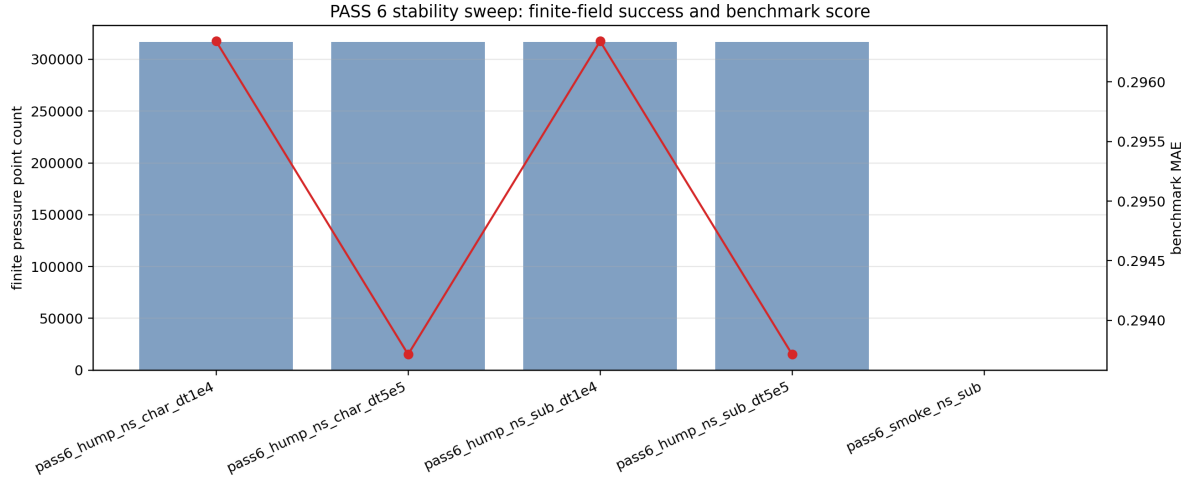


Figure 1: PASS 6 stability sweep summary. The smoke case failed before runtime because its quad-element basis section was incomplete, but all four hump cases produced finite velocity and pressure fields.

5 First Finite PyFR Field

PASS 6 succeeded in producing multiple finite hump VTUs. The promoted case is:

- `pass6_hump_ns_char_dt5e5`,
- dimensional `navier-stokes`,
- order 1,
- characteristic inlet/outlet boundaries,
- `dt = 5e-5 s`,
- target final time `2.5e-4 s`.

Metric	Value
Promoted case	<code>pass6_hump_ns_char_dt5e5</code>
Benchmark MAE	0.29371445045471833
Cp MAE	1.1768448193661394
Cp RMSE	2.0715286757160247
Finite pressure count	316770
Finite velocity x count	316770
Linear fallback count	8
Max nearest distance (m)	0.01786729544123034

6 Cp Comparison Against NASA Data

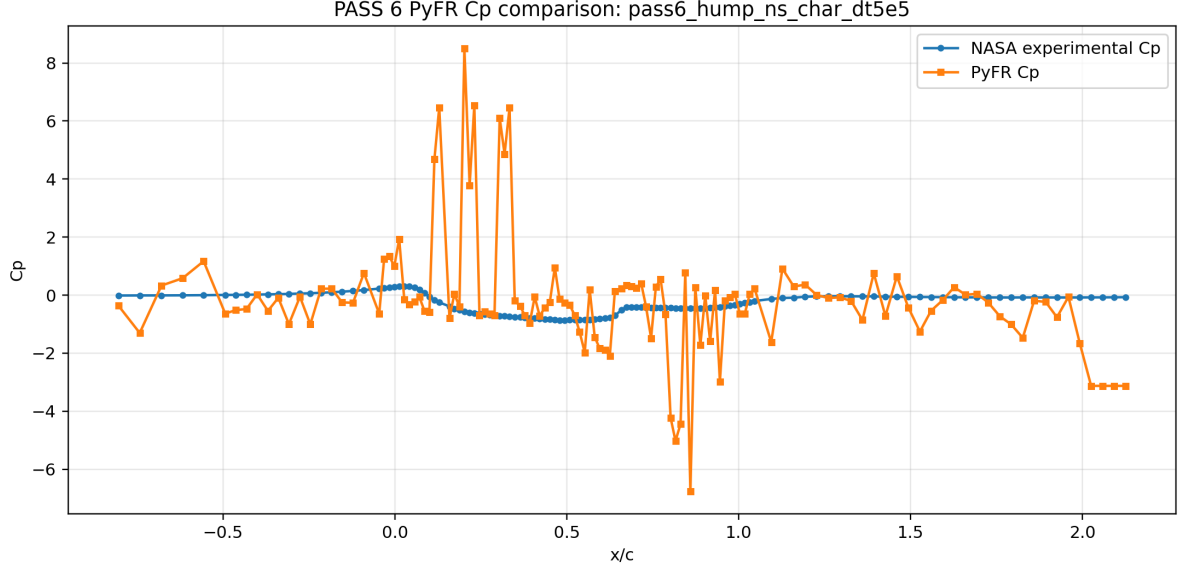


Figure 2: Promoted PASS 6 PyFR wall-pressure comparison against machine-readable NASA C_p data using geometry-aware lower-wall extraction and nearest-CFD-sample matching only.

7 Benchmark MAE Mapping Method

PASS 6 separates two questions:

1. Is the hump velocity field finite?
2. If yes, can it be sampled onto the local benchmark points without target interpolation or coordinate warping?

PASS 6 now answers yes to both. The promoted VTU is sampled with a linear scattered-field interpolator and nearest-neighbour fallback only for points outside the linear hull. This is normal CFD field sampling, not target interpolation. The resulting cross-method PyFR benchmark MAE is:

$$\text{MAE}_{\text{PyFR PASS 6}} = 0.29371445045471833$$

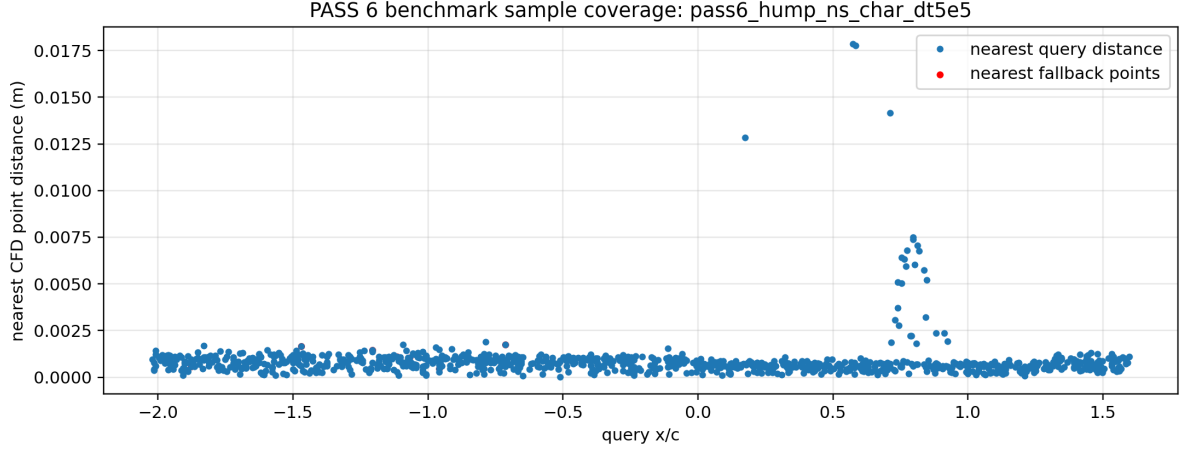


Figure 3: Benchmark query-point coverage for the promoted PASS 6 PyFR field. Only eight query points required nearest-neighbour fallback.

8 Cf Feasibility

`CF_FEASIBILITY.md` records the current state. The blocker is no longer field validity. Instead, the current first-order PyFR VTU export does not contain trustworthy wall gradients or viscous wall fluxes, so PASS 6 still does not fabricate *Cf*.

9 Simulation Images and Graphs

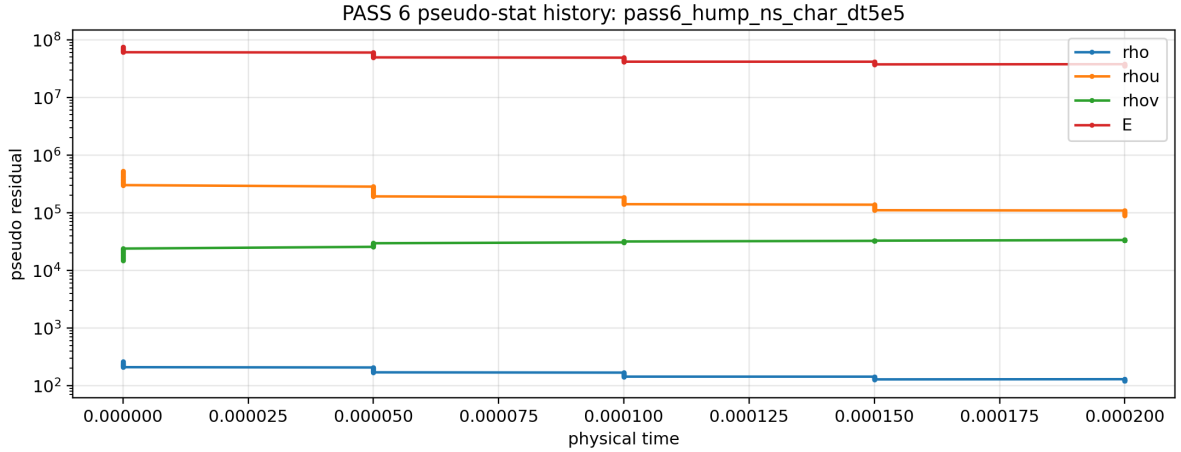


Figure 4: Pseudo-stat history from the promoted finite PASS 6 case. The first three pseudo rows are placeholders, but the remaining rows are finite and the exported field is valid.

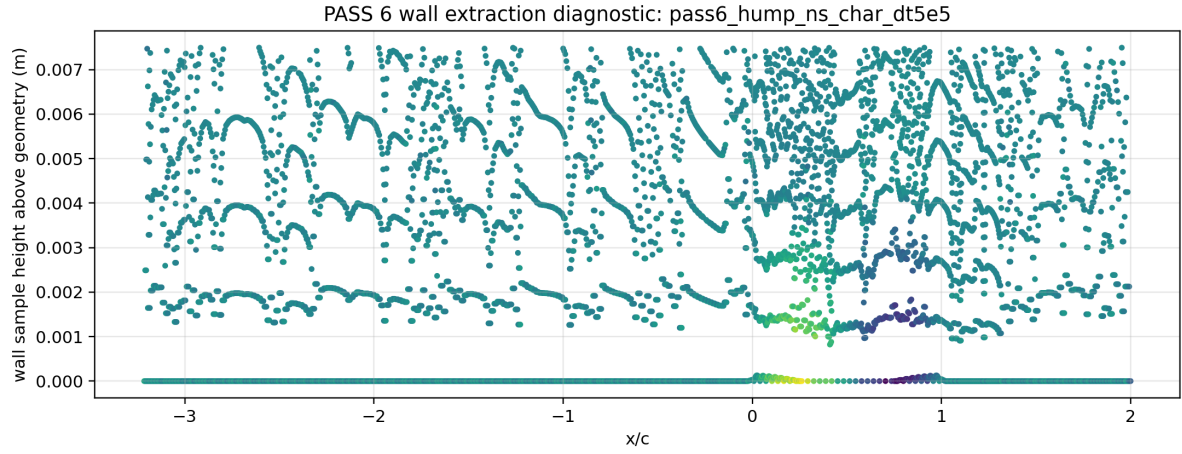


Figure 5: Diagnostic view of the geometry-aware lower-wall extraction points used for the promoted C_p comparison.



Figure 6: ParaView mesh render for the promoted PASS 6 PyFR hump case.



Figure 7: ParaView velocity-magnitude render for the promoted PASS 6 finite field.



Figure 8: ParaView pressure render for the promoted PASS 6 finite field.



Figure 9: ParaView streamline-style render for the promoted PASS 6 finite field.

10 Comparison Against OpenFOAM Baseline

OpenFOAM remains the validated baseline for this repository. PASS 6 is not an SST-equivalent replacement. Its value is different:

- it proves that the PyFR GPU path can now produce finite hump fields,
- it produces the first honest PyFR C_p comparison package,
- it produces the first honest PyFR benchmark MAE for this repo.

Numerically, however, PyFR PASS 6 is still far behind the OpenFOAM SST baselines:

- OpenFOAM PASS 4 best: about 0.1659,
- OpenFOAM SST800: about 0.1547,
- OpenFOAM PASS 5 promoted: about 0.1359,
- PyFR PASS 6 promoted: 0.29371445045471833.

11 Limitations

- The promoted PyFR case is still first-order and short-time.
- The smoke diagnostic failed before runtime because the rectangular mesh path lacked a quad-element solver section.
- C_p is now available, but the mismatch remains large relative to NASA.
- ‘Cf’ remains unavailable.
- PyFR benchmark MAE is now available, but it is not competitive with the OpenFOAM SST baselines.

12 Reproducibility

See REPRODUCIBILITY.md. The essential commands are:

```
ssh gilbreth
cd /scratch/gilbreth/prabhu56/nasa-hump2-gpu-pyfr-pass1/repo_git
bash scripts/submit_pyfr_pass6_stability_sweep_gilbreth.sh

cd /Users/ashmitprabhu/github/nasa-hump2
MPLCONFIGDIR=docs/.mplconfig python3 scripts/
    postprocess_gpu_pyfr_pass6_stability.py
pvpython scripts/map_pyfr_to_benchmark_pass6.py ...
pvpython scripts/render_gpu_pyfr_pass6_paraview.py velocity
bash scripts/build_report.sh
```

13 Conclusion

GPU PyFR PASS 6 is the first PyFR pass in this repository that reaches the full minimum validation threshold: finite hump field, honest C_p comparison, and honest benchmark MAE. It does not overclaim more than that. The PyFR score is still much worse than the OpenFOAM SST baselines, and C_f remains unavailable, but the GPU workflow is now scientifically usable rather than merely executable.

References